L05 – Seismic Data Formats

1. Types of data formats used in Seismology

SEED (Standard for the Exchange of Earthquake data)

1. It is used for archival and related metadata.
2. Contain everything e.g. Metadata, response file, pole zero files etc.
3. Contained more memory space.

MSEED (MiniSEED)

1. It's part of SEED.
2. contain only waveform data. There is no station and channel metadata included.
3. Take less memory space.

Dataless SEED

1. It contains only station and channel metadata.

SEISAN

1. The SEISAN format is probably a best choice because easily used on all computer platforms.
2. The data format can only process in SEISAN software.

ASCII

1. This is a simple text-based data format and not well suited for serious data processing.

GeoCSV

1. This is very similar to the Simple ASCII format, but includes some additional metadata

1. `

2. Seismic Analysis Code (SAC)

2.1 Overview

1. SAC stands for "Seismic Analysis Code". It was originally developed at Lawrence Livermore National Laboratory to analyze data in time series, especially seismic data. It is one of the most widely used data formats and programs in the seismological research community.

3. Recently, SAC source code becomes publicly available to IRIS membership institutions. Related website for requesting a copy of the SAC software is http://ds.iris.edu/ds/nodes/dmc/forms/sac/

4. The best reference for SAC can be found from the IRIS website: SAC Software Manual , and SAC Tutorial Guide for New Users . Other useful links about SAC are SAC Guide by Charles Ammon at Penn State , SAC Tutorial by Charles Langston at CERI , and SAC Tutorial by Brian Savage at URI.

2.3 Installation

Note: All the commands are in italics and underlined

1. Come to your home directory by typing: cd

2. Create a new directory "SAC": mkdir SAC_LECTURE

3. Go into SAC_LECTURE: cd SAC_LECTURE; Create another directory: mkdir sac

4. Keep "sac-101.6a-linux_x86_64.tar.gz" in directory "sac": use copy command "cp"

5. Change your directory to sac: cd sac

6. Extract "sac": tar xvf sac-101.6a-linux_x86_64.tar.gz

7. Find your path: pwd ; and copy your path

8. Open ".bashrc" using: vi ~/.bashrc ; and keep the following lines in .bashrc and save it

   #SAC
   export SACHOME=paste your copied path here
   . ${SACHOME}/bin/sacinit.sh

9. Now change your directory to bin: cd bin

10. Open "sacinit.sh": vi sacinit.sh

11. Go on to line number 17 and replace right hand side of "=" with the path that you copied in step. Don't forget to save it.

12. Open new terminal and type: sac; you should see the followings:
    SEISMIC ANALYSIS CODE [11/11/2013 (Version 101.6a)]
    Copyright 1995 Regents of the University of California
    SAC>

if you can see this; then you have successfully installed sac.

2.4. Read/Write/Plot seismic data in SAC

Next, we will read some seismograms, and play with it. Let's suppose we would like to perform some operations on an earthquake file 'BV.Z' in SAC as follows:

```
% sac
SAC> r BV.z              # read the data file
SAC> p                   # plot a single data
SAC> r BV.?              # read all three component data recorded at station BV
SAC> p1                  # plot multiple data in one page
SAC> xlim 0 7            # zoom in at 0 - 7 s relative to the waveform start time
SAC> p1                  # you have to plot it again to reflect the change
SAC> xlim a -2 t0 +4     # zoom in at 2 s before P, and 4 s after S arrivals
SAC> ylim all            # enforce that y axis has the same scale
SAC> p1                  # you have to plot it again to reflect the change
SAC> bd sgf              # Begins plotting to the SAC Graphics File device driver
SAC> p1                  # no X graphic is shown. But a file named f001.sgf is created
SAC> sgftops f001.sgf BV_3c.ps # Convert the SGF file into a postscript file
SAC> quit

% gs BV_3c.ps            # ghostview your postscript file
```

To write SAC data back to your disk, you have the following three options. See below for examples.

```
%sac
SAC> r BV.z              # read the data file
SAC> p1                  # plot it out
SAC> rmean               # remove the mean
SAC> w over              # overwrite the data back to the disk (using the original name BV.z)
```

SAC> w append .sac # Write the data back to the disk using a new name BV.z.sac

SAC> w temp.sac # Write the data back to the disk using a new name temp.sac

SAC> quit

% rm -rf temp.sac BV.z.sac # clean up what just created.

To learn the syntax of each command, use "help command".

You can use short version of a command. For example, "p" is the same as "plot", "w" is the same as "write".

2.5 SAC Header

SAC header consists of important information about the data, such as sampling interval, start time, length, station location, event location, components, phase arrival, etc. There parameters will be used by various SAC command to process the data. It is important to keep your header information complete and updated.

To list the SAC header information, you can open the data in SAC, and use lh (listhdr) command. To list a specific header only, use lh header_name (e.g., lh delta). To go back to the default listing (all headers), use lh default.

% sac # launch sac again

SAC> r BV.z # read the data again

SAC> lh # list the SAC header

FILE: BV.z - 1

NPTS = 2409 # number of data points

B = -9.676000e+00 # begin time

E = 1.440400e+01 # end time

IFTYPE = TIME SERIES FILE # file type

LEVEN = TRUE # evenly sampled time series

DELTA = 1.000000e-02 # time increment

IDEP = VELOCITY (NM/SEC) # physical unit of the data

DEPMIN = -2.073471e+04 # minimum amplitude

```
 DEPMAX = 1.584818e+04     # maximum amplitude
 DEPMEN = 5.137106e+01     # mean amplitude
OMARKER = 0.317            # event origin marker
AMARKER = 2.165            # first arrival (P) marker
T0MARKER = 3.509           # t0 (S) marker
 KZDATE = NOV 20 (324), 1999          # reference date
 KZTIME = 00:12:55.523                # reference time
 IZTYPE = GMT DAY                     # type of reference time
  KSTNM = BV                          # station name
  CMPAZ = 0.000000e+00                # component azimuth relative to north
 CMPINC = 0.000000e+00                # component "incidence angle" reletive to the vertical
   STLA = 4.075520e+01                # station latitude
   STLO = 3.101490e+01                # station longitude
   STEL = 2.470000e+02                # station elevation
   STDP = 0.000000e+00                # station depth below surface (meters)
   EVLA = 4.079930e+01                # event latitude
   EVLO = 3.100330e+01                # event longitude
   EVDP = 8.150000e+00                # event depth
   DIST = 4.994444e+00                # source receiver distance in km
     AZ = 1.686886e+02                # azimuth
    BAZ = 3.486961e+02                # back azimuth
  GCARC = 4.492941e-02                # great circle distance
 LOVROK = TRUE                        # TRUE if it is okay to overwrite this file on disk
  USER7 = 0.000000e+00                # User defined time picks
  USER8 = 0.000000e+00                # User defined time picks
  NVHDR = 6                # Header version number. Current value is the integer 6.
  SCALE = 1.000000e+00     # Multiplying scale factor for dependent variable
```

NORID = 0 # Origin ID (CSS 3.0)

NEVID = 0 # Event ID (CSS 3.0)

NWFID = 2 # Waveform ID (CSS 3.0)

LPSPOL = FALSE # TRUE if station components have a positive polarity

LCALDA = TRUE # TRUE if DIST, AZ, BAZ, and GCARC are to be calculated from station and event coordinates

KCMPNM = EPZ_01 # Component name

MAG = 2.310000e+00 # Event magnitude

A detailed description of the SAC data file format, including the header, can be found at the SAC2000 User's manual http://www.adc1.iris.edu/files/sac-manual/sac_manual.pdf

Another way to list SAC header is to use a C program saclst (originally written by Professor Lupei Zhu at SLU. The binary code is in /usr/local/geophysics/bin.

% saclst

Usage: saclst header_lists f file_lists

% saclst evla evlo stla stlo f BV.? # list the SAC header evla evlo stla stlo

BV.e 40.7993 31.0033 40.7552 31.0149

BV.n 40.7993 31.0033 40.7552 31.0149

BV.z 40.7993 31.0033 40.7552 31.0149

2.6 Other basic tools available in SAC

Many basic tools are available in the SAC program. Below I will show a few examples of these tools and functions. A complete list of the functions can be found online at SAC Commands Listing: Functional .

Rotate a pair of data.

We can easily rotate a pair of data components through an angle in SAC using the "rotate" command. Each pair must have the same station name, event name, and sampling rate. The most common practice is seismology is to rotate two horizontal component seismograms to be along and perpendicular to the "great circle path" (i.e., radial and transverse component). This means that CMPAZ must be defined and that CMPINC must be 90 degrees. See below for an example of rotating two horizontal seismograms at Kota Tinggi in Malaysian and generated by the 2006/07/17 M7.7 JAVA Tsunami earthquake.

```
% sac
SAC> r KOM.BHN.SAC KOM.BHE.SAC          # read two horizontal component seismograms
SAC> rotate to GCP                      # rotate to the "great circle path"
SAC> w KOM.BHR.SAC KOM.BHT.SAC          # save the rotated data (first radial, second
                                          transverse)
SAC> r KOM.BHZ.SAC KOM.BHR.SAC KOM.BHT.SAC # read three-component seismograms
SAC> qdp off                            # turn off "quick and dirty plot"
SAC> xlim 100 1000                      # zoom in
SAC> p1                                 # plot them out, try to identify Love and Rayleigh waves
SAC> cut 550 650                        # prepare the data window for Rayleigh wave
SAC> r KOM.BHZ.SAC KOM.BHR.SAC          # read the vertical and radial component only
SAC> xlim off                           # turn off the zoom in time window
SAC> p2                                 # plot them on top of each other
SAC> ppm                                # plot the partical motion
SAC> quit                               # done
```

Phase pick.

Phase picking can be easily done in SAC via automatic phase picker "apk", or manually phase picker "ppk".

```
% sac
SAC> r BV.?             # read three component data
SAC> qdp off            # turn off quick dirty plot
SAC> p1                 # plot the data
SAC> apk                # use the auto picker
SAC> p1                 # see the auto picker result
SAC> ppk                # use the manual picker
```

cheat sheet,

zoom in: type "x" to define left side time window,

followed by a left click of mouse to define the right side time window

zoom out: type "o"

p arrival: type "a", or "p" at the time

where you think the p wave arrival in.

s arrival: type "t0", or "s" at the time

where you think the s wave arrival in.

quit: type "q"

```
SAC> ppk p 3 m on          # pick all 3 component together with the same time
SAC> wh                    # save the header information, be careful
```

Compute cross correlation.

Below is an example of how to compute cross correlation for waveforms generated a group of repeating earthquakes and recorded at station CCO in northern California.

```
%sac
SAC> cut t0 -0.5 t0 +2.5   # cut the data 0.5 s before and 2.5 s after the s arrival
SAC> r CCO*.EHZ     # read all the waveforms
SAC> p2             # plot the data on top of each other
SAC> correlate      # compute waveform cross correlation with respect to the first data
SAC> xlim -0.5 0.5  # zoom in the cross-correlation functions
SAC> p1             # plot out the cross-correlation functions
```